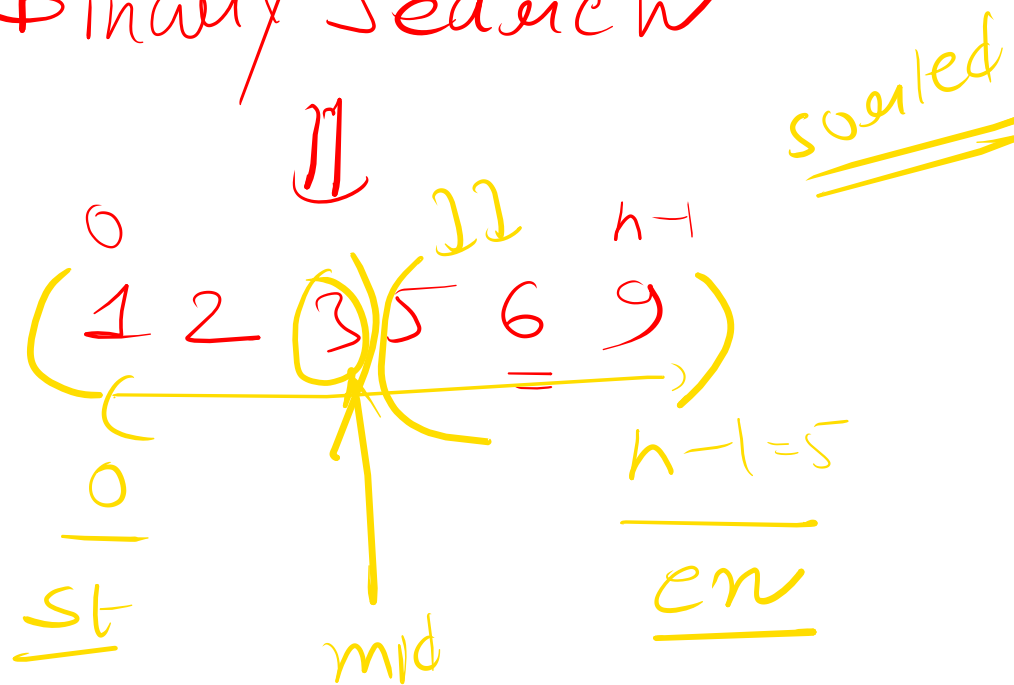


any kind
 LS
 0 1 5 2 3 4 6
 n-1

st ... en
 ↓
 search space

→ K=6

Binary search

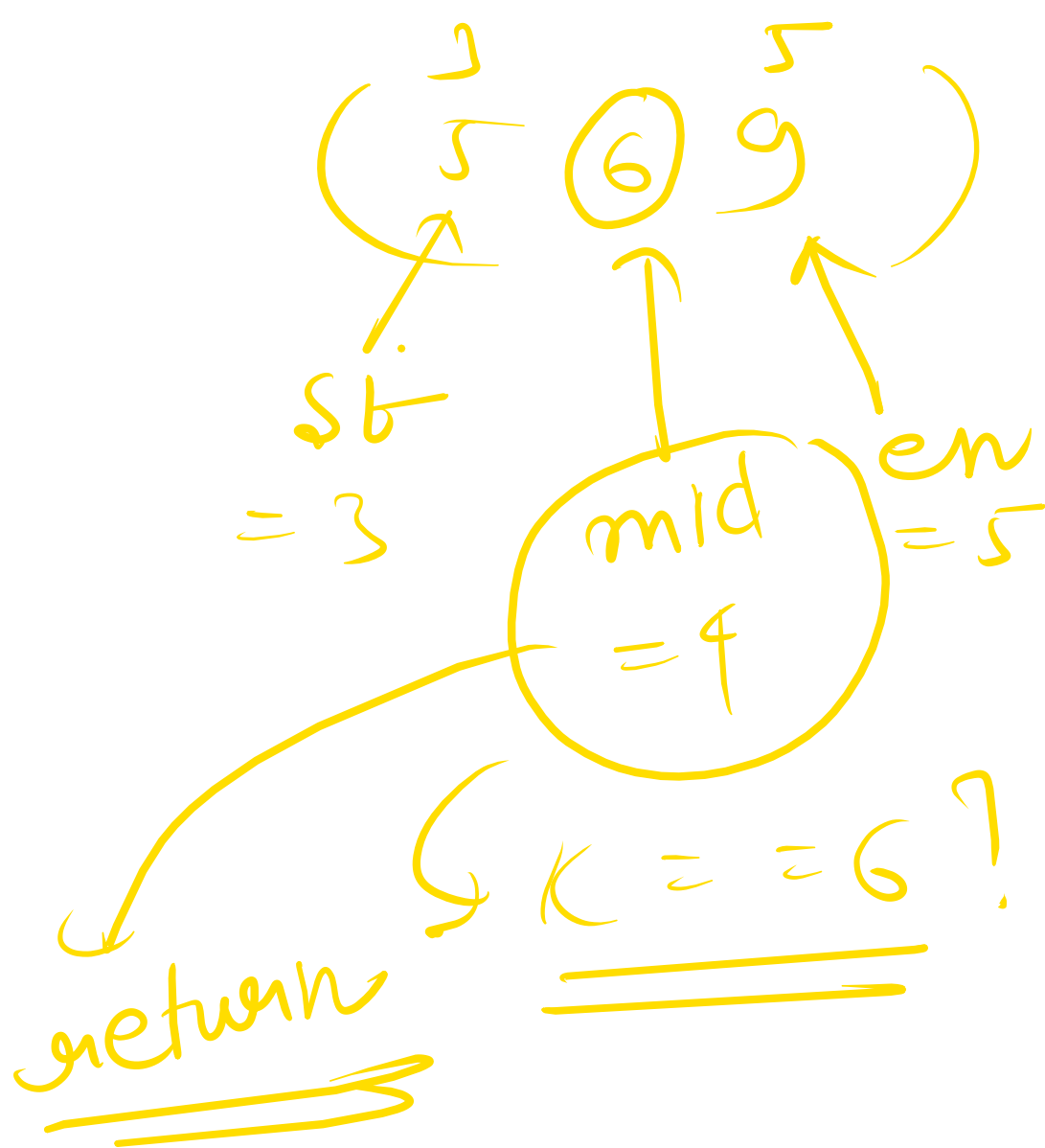


$$= \frac{(st + en)}{2} = 2$$

(st ← mid) , greatest < K

K ≠ 3
 K < 3?
 ↑ K > 3 ↑?

$$\text{mid} = \frac{3+5}{2} = 4$$



$$\underline{K=1}$$

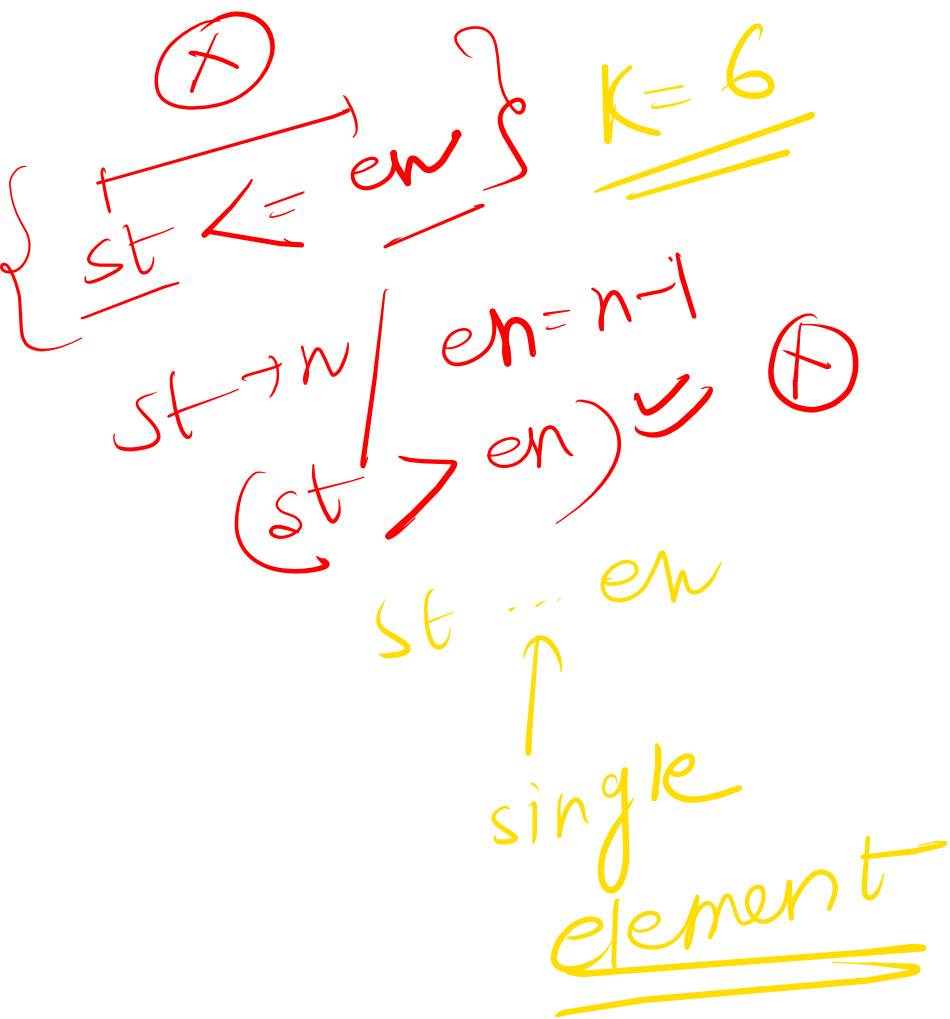
$(1\ 5\ 6\ 7)\ 8\ 9\ 10\ 11\ 12$
 $st=0$ $en=8$ $\frac{mid}{8} > K$
 en $\underline{mid=9}$

$$\underline{K=1}$$

$(mid \dots en)$
 $\uparrow$
 $smallest$
 $> \underline{K}$

$(1\ 5\ 6\ 7) \rightarrow 5 > \underline{K}$
 $st \rightarrow 0$ $mid \rightarrow 3$ $en \rightarrow 3$

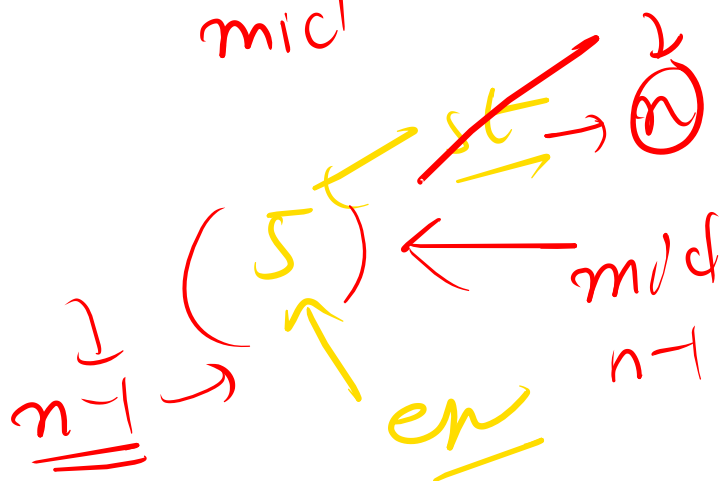
$(1) \leftarrow \begin{matrix} st \\ mid \\ en \end{matrix}$ $1 == K$
 $\rightarrow \underline{mid=6}$
 $\underline{ans}$



$$3 < \underline{K}$$



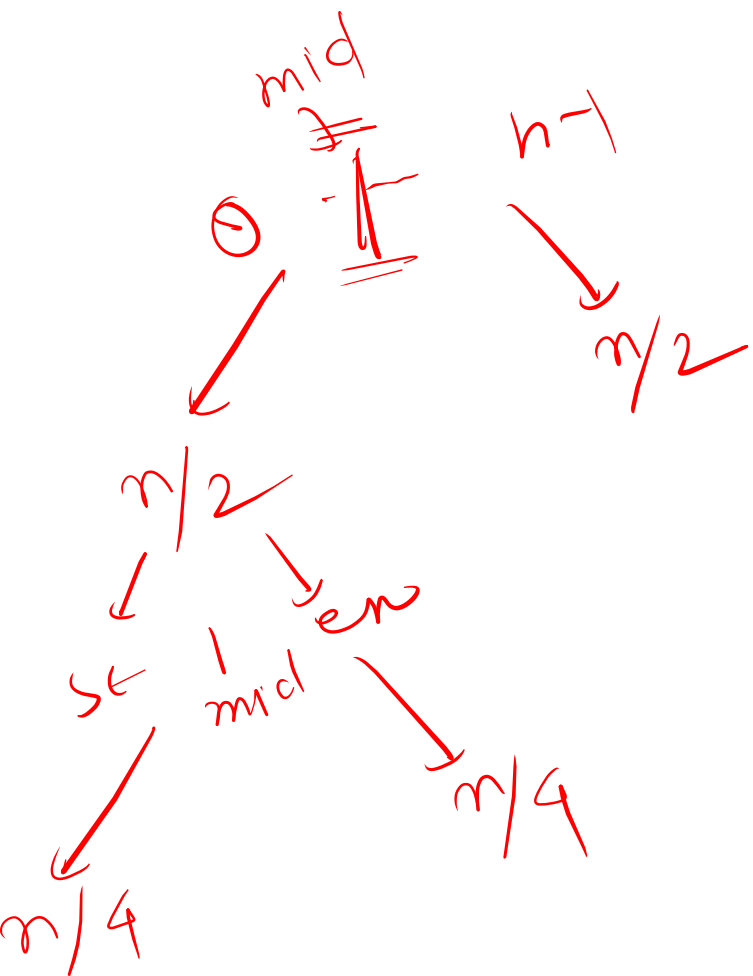
$$\frac{3+4}{2} = 3$$



$$4 < \underline{K}$$

$$5 < \underline{K}$$





Time complexity.

$\rightarrow \underline{n} \rightarrow \underline{mid}$

$\rightarrow \underline{n/2} \rightarrow \underline{mid}$

$\rightarrow \underline{n/4} \rightarrow \underline{mid}$

$\rightarrow \underline{n/8} \rightarrow \underline{mid}$

$\rightarrow \dots \rightarrow \underline{1} \rightarrow \underline{mid}$

1

$\equiv \underline{\underline{O(\log(n))}}$

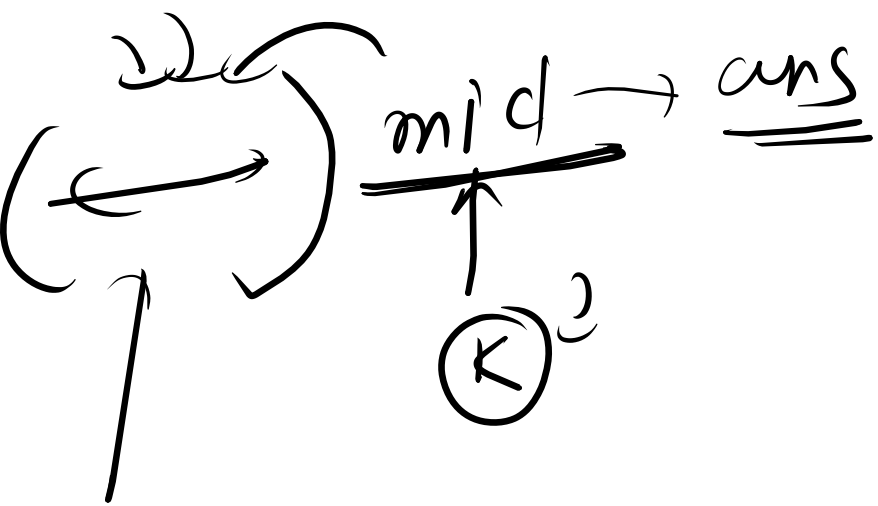
$\downarrow h=9$
 $\downarrow 9^2 > x$
 $ans = 4$
 $4^2 \leq 18$

$x = 18$
 $9^2 \leq 18 \leftarrow 0$
 $\times$
 $4^2 \leq 18 \checkmark$
 $6^2 \leq 18 \times$
 0
 9
 4
 mid
 5
 6
 8
 mid

5
 mid
 en
 $5^2 \leq 18 \times$
 $en = 4$
 $st = 5$
 $st > en \times$

sorted $\rightarrow$ 1 1 2 3 3 4 4 4 9 $\xrightarrow{K=4}$ 5
first occ.

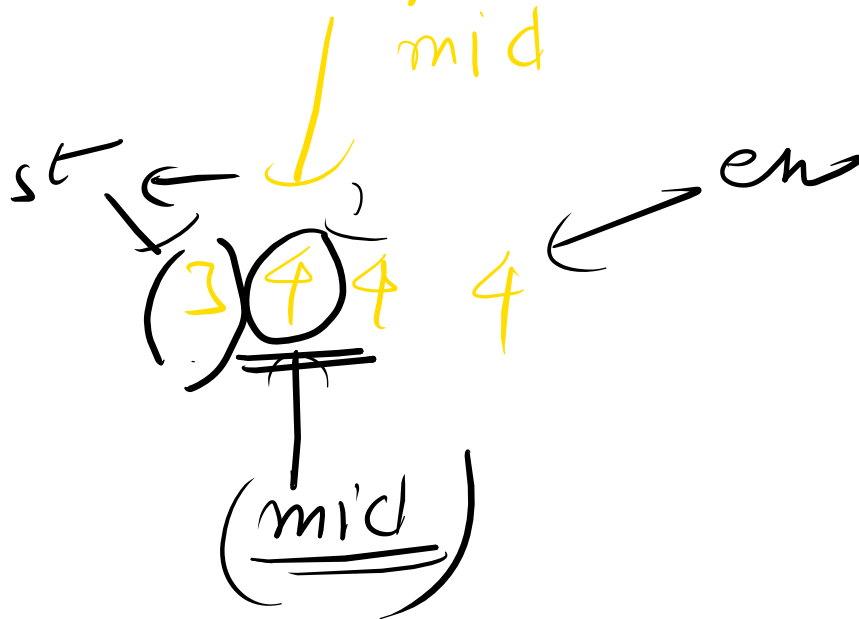
linear 0 i
 $\rightarrow$ 1 1 2 3 3 4 4 4 ans
K $\rightarrow$ $O(n)$



$st > en$
 $\otimes$

st en
 1 1 2 3 3 4 4 4

$3 < K$



$4 = K$

$ans = 5$

st en
 $mid \rightarrow 3$
 $\otimes$

$st = mid + 1$
 4 5 / $en = 4$

$3 < K$

first $\rightarrow$ 2 $\frac{\log(n)}{\log(n)}$ $\leftarrow$ $\overset{0}{(1\ 2\ 2\ 2)} \textcircled{2} 2\ 2\ 2\ 3\ \overset{9}{4}$
 st en

$k=2$

$\text{st} \text{ } \text{en}$
 $\textcircled{+}$

$2 == k$

ans = 4 $\textcircled{1}$

mid $\textcircled{4}$

st en
 $\overset{0}{(1)} \textcircled{2} 2\ 2\ 3$

$1 < k$

$\text{st} = \text{mid} + 1$
 $\text{en} = 0$ $\textcircled{1}$ mid

$\text{mid} = \textcircled{1}$

$2 == k$

else if (arr[mid] > k) $\left\lceil \frac{\text{arr}[k]}{n} \right\rceil$
 eh = mid - 1;

else
 st = mid + 1;

}
return ans;

st = 0 ans = -1
eh = n - 1

while (st <= eh) {

 mid = (st + eh) / 2;

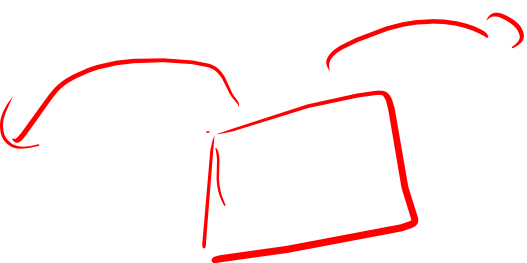
 if (arr[mid] == k) {

 ans = mid;

 eh = mid - 1; // st = mid + 1

 }

mid $\rightarrow$ $O(\log(n))$



BS

$\frac{n}{1}$
 $\downarrow$
 $\frac{n}{2} \rightarrow \underline{O(\log(n))}$
 $\downarrow$
 $\frac{n}{4}$
 $\downarrow$
 $\frac{n}{8}$
 $\vdots$
 $\textcircled{1}$

$\Theta(n)$
 $\swarrow$
 $O(\log(n))$

$K=3$

sorted
BS

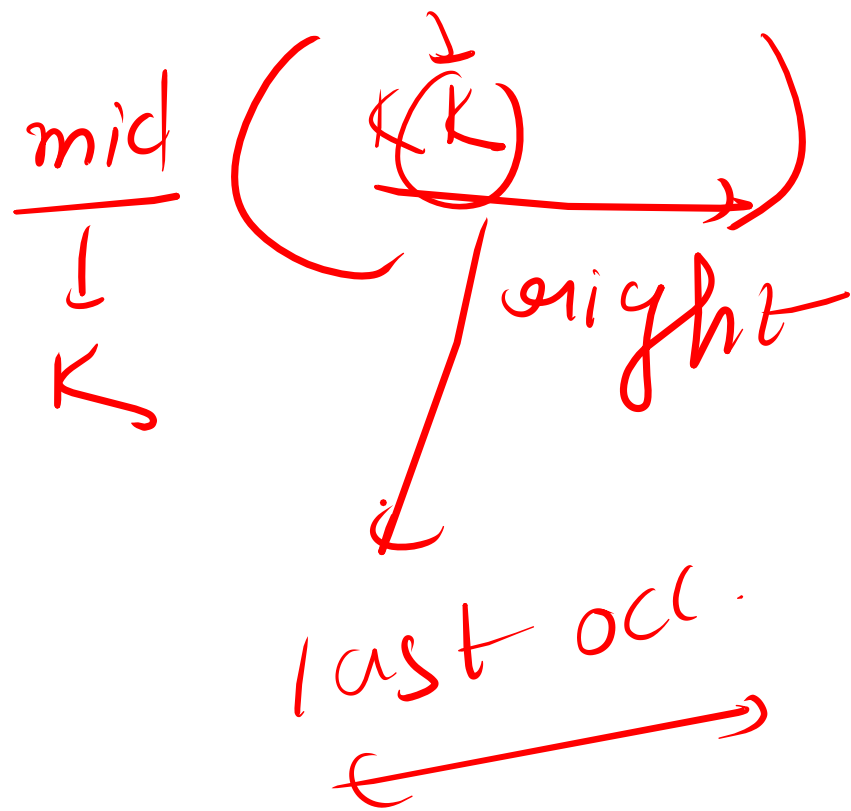
0 $h-1$
 1 1 2 2 2 2 2 3 3
 i
 $ans = i$

$3 == K$
 $ans = 7$
8

$2 < K$
 $2 < K$

st st en
 1 1 2 2 (2) (2) (2) (3 3)
mid mid

st en
 (3) (3)
mid
 (x) st $mid = 8$
en = 8



$$\begin{array}{c}
 n \\
 | \\
 n/2 \\
 | \\
 n/4 \\
 \vdots \\
 1
 \end{array}
 \equiv \underline{\underline{O(\log n)}}$$